

ESO208 COMPUTATIONAL METHODS FOR ENGINEERS

ASSIGNMENT 1

MAXIMUM MARKS: 30

DUE DATE: 14 AUGUST 2026

Please upload hand-written solutions on Gradescope. Typed solutions except codes and plots will not be graded. Please upload single PDF file for hand-written part.

Codes used in the assignment are to be uploaded separately.

Q.1. [10 marks] The average daily inflow rate of water into a dam for 5 days is listed below

Day	Inflow (m^3/s)
1	1.2685×10^1
2	2.1345×10^1
3	1.8432×10^1
4	9.0172×10^0
5	1.3210×10^1

- (a) If there is no water released from the dam, what will be the increase in dam storage by the end of the 5-day period? [2]
- (b) Calculate the chop-off error in (a) if only 3 significant digits of inflow measurements are used for calculation. [4]
- (c) Calculate the round-off error in (a) if only 3 significant digits of inflow measurements are used for calculation. [4]

Q.2. [10 marks] The saturation vapor pressure of the atmosphere e_s denotes the pressure exerted by water vapor when the air is fully saturated. e_s in Pascals can be calculated using the air temperature T in $^{\circ}\text{C}$ as

$$e_s = 611 \exp\left(\frac{17.27T}{237.3 + T}\right)$$

- (a) What will be the error in e_s given an error ΔT in temperature measurement? [5]
- (b) If $\Delta T = \pm 0.1^{\circ}\text{C}$, calculate the error in e_s at $T = 20^{\circ}\text{C}$ and $T = 30^{\circ}\text{C}$. [5]

Q.3. [10 marks] The position of an object y as a function of time t is given by

$$y(t) = \sin\left(\frac{2\pi}{T}t\right)$$

where $0 \leq t \leq T$.

- (a) Calculate the condition number of the function $y(t)$ for $0 \leq t \leq T$ (manually). [3]
- (b) Plot the condition number of the function $y(t)$ for $0 \leq t \leq T$ (show the code). [3]
- (c) Based on the plot in (b), at what values of t , the relative error in y will be highest for a given relative error in t . Please give reason. [4]